

公式推导

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1 两球三弹簧谐振子

如图1,该物理模型由两个完全相同的小球(质量为 m)与三个弹簧(劲度系数 $k = m\omega^2$)连接,以及两面质量无穷大的刚体墙构成。

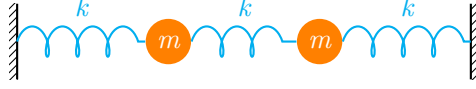


图 1: 两球三弹簧谐振子

该系统的拉格朗日量如下表示。

$$\begin{cases} L = T - V \\ T = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 \\ V = \frac{1}{2}m\omega^2 x_1^2 + \frac{1}{2}m\omega^2 x_2^2 + \frac{1}{2}m\omega^2 (x_1 - x_2)^2 \end{cases} \quad (1)$$

此时有

$$\begin{cases} \frac{\partial}{\partial x_1}L - \frac{d}{dt}\left(\frac{\partial}{\partial \dot{x}_1}L\right) = 0 \\ \frac{\partial}{\partial x_2}L - \frac{d}{dt}\left(\frac{\partial}{\partial \dot{x}_2}L\right) = 0 \end{cases} \implies \frac{d^2}{dt^2} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \omega^2 \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \quad (2)$$

对角化即有

$$\frac{d^2}{dt^2} \begin{pmatrix} \frac{1}{\sqrt{2}}(x_1 + x_2) \\ \frac{1}{\sqrt{2}}(x_1 - x_2) \end{pmatrix} + \omega^2 \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}(x_1 + x_2) \\ \frac{1}{\sqrt{2}}(x_1 - x_2) \end{pmatrix} = 0 \quad (3)$$

即有

$$x_1 + x_2 = C_{c1} \cos(\omega t) + C_{s1} \sin(\omega t) \quad (4)$$

$$x_1 - x_2 = C_{c3} \cos(\sqrt{3}\omega t) + C_{s3} \sin(\sqrt{3}\omega t) \quad (5)$$

其中 C_{c1} 、 C_{s1} 、 C_{c3} 、 C_{s3} 均为与初始状态相匹配的常数。

初始状态与常数的关系如下

$$\begin{cases} x_1(t=0) = X_1 \\ x_2(t=0) = X_2 \\ \dot{x}_1(t=0) = V_1 \\ \dot{x}_2(t=0) = V_2 \end{cases} \implies \begin{cases} C_{c1} = X_1 + X_2 \\ C_{c3} = X_1 - X_2 \\ C_{s1} = \frac{V_1 + V_2}{\omega} \\ C_{s3} = \frac{V_1 - V_2}{\sqrt{3}\omega} \end{cases} \quad (6)$$

2 双摆系统

如图2，两个完全相同的小球通过长度相同的不可延伸硬杆相连接，构成双摆。该系统具备混沌的特点，但并非所有初态均对应混沌状态。

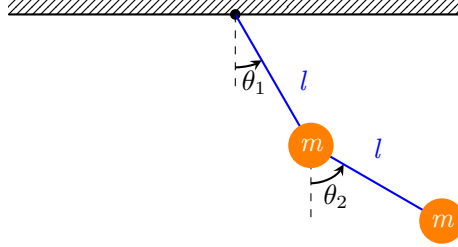


图 2: 双摆系统

2.1 物理过程

该体系拉格朗日量如下所示

$$\begin{cases} L = T - V \\ T = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 + \frac{1}{2}m\dot{y}_1^2 + \frac{1}{2}m\dot{y}_2^2 \\ V = -mgy_1 - mgy_2 \end{cases} \quad (7)$$

根据双摆的规则，有

$$\begin{cases} x_1 = l \sin(\theta_1) \\ x_2 = l \sin(\theta_1) + l \sin(\theta_2) \\ y_1 = l \cos(\theta_1) \\ y_2 = l \cos(\theta_1) + l \cos(\theta_2) \end{cases} \Rightarrow \begin{cases} \dot{x}_1 = l \cos(\theta_1) \dot{\theta}_1 \\ \dot{x}_2 = l \cos(\theta_1) \dot{\theta}_1 + l \cos(\theta_2) \dot{\theta}_2 \\ \dot{y}_1 = -l \sin(\theta_1) \dot{\theta}_1 \\ \dot{y}_2 = -l \sin(\theta_1) \dot{\theta}_1 - l \sin(\theta_2) \dot{\theta}_2 \end{cases} \quad (8)$$

代入则有

$$\begin{cases} T = \frac{1}{2}ml^2 \left(\cos^2(\theta_1) \dot{\theta}_1^2 + \left(\cos(\theta_1) \dot{\theta}_1 + \cos(\theta_2) \dot{\theta}_2 \right)^2 + \sin^2(\theta_1) \dot{\theta}_1^2 + \left(\sin(\theta_1) \dot{\theta}_1 + \sin(\theta_2) \dot{\theta}_2 \right)^2 \right) \\ V = -2mgl \cos(\theta_1) - mgl \cos(\theta_2) \end{cases} \quad (9)$$

即

$$\begin{cases} T = \frac{1}{2}ml^2 \left(2\dot{\theta}_1^2 + \dot{\theta}_2^2 + 2 \cos(\theta_1 - \theta_2) \dot{\theta}_1 \dot{\theta}_2 \right) \\ V = -2mgl \cos(\theta_1) - mgl \cos(\theta_2) \end{cases} \quad (10)$$

$$L = \frac{1}{2}ml^2 \left(2\dot{\theta}_1^2 + \dot{\theta}_2^2 + 2 \cos(\theta_1 - \theta_2) \dot{\theta}_1 \dot{\theta}_2 \right) + 2mgl \cos(\theta_1) + mgl \cos(\theta_2) \quad (11)$$

固有

$$\frac{\partial L}{\partial \theta_1} = \frac{1}{2}ml^2 \cdot -2\sin(\theta_1 - \theta_2)\dot{\theta}_1\dot{\theta}_2 - 2mgl\sin(\theta_1) \quad (12)$$

$$\frac{\partial L}{\partial \theta_2} = \frac{1}{2}ml^2 \cdot 2\sin(\theta_1 - \theta_2)\dot{\theta}_1\dot{\theta}_2 - mgl\sin(\theta_2) \quad (13)$$

$$\frac{\partial L}{\partial \dot{\theta}_1} = \frac{1}{2}ml^2 (4\dot{\theta}_1 + 2\cos(\theta_1 - \theta_2)\dot{\theta}_2) \quad (14)$$

$$\frac{\partial L}{\partial \dot{\theta}_2} = \frac{1}{2}ml^2 (2\dot{\theta}_2 + 2\cos(\theta_1 - \theta_2)\dot{\theta}_1) \quad (15)$$

则

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) = \frac{1}{2}ml^2 (4\ddot{\theta}_1 + 2\cos(\theta_1 - \theta_2)\ddot{\theta}_2 - 2\sin(\theta_1 - \theta_2)\dot{\theta}_1\dot{\theta}_2 + 2\sin(\theta_1 - \theta_2)\dot{\theta}_2^2) \quad (16)$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_2} \right) = \frac{1}{2}ml^2 (2\ddot{\theta}_2 + 2\cos(\theta_1 - \theta_2)\ddot{\theta}_1 + 2\sin(\theta_1 - \theta_2)\dot{\theta}_1\dot{\theta}_2 - 2\sin(\theta_1 - \theta_2)\dot{\theta}_1^2) \quad (17)$$

代入

$$\begin{cases} \frac{\partial}{\partial \theta_1} L = \frac{d}{dt} \left(\frac{\partial}{\partial \dot{\theta}_1} L \right) \\ \frac{\partial}{\partial \theta_2} L = \frac{d}{dt} \left(\frac{\partial}{\partial \dot{\theta}_2} L \right) \end{cases} \quad (18)$$

有

$$\begin{cases} \frac{1}{2}ml^2 (4\ddot{\theta}_1 + 2\cos(\theta_1 - \theta_2)\ddot{\theta}_2 + 2\sin(\theta_1 - \theta_2)\dot{\theta}_2^2) + 2mgl\sin(\theta_1) = 0 \\ \frac{1}{2}ml^2 (2\ddot{\theta}_2 + 2\cos(\theta_1 - \theta_2)\ddot{\theta}_1 - 2\sin(\theta_1 - \theta_2)\dot{\theta}_1^2) + mgl\sin(\theta_2) = 0 \end{cases} \quad (19)$$

考虑 $\omega = \sqrt{\frac{g}{l}}$, 有

$$\begin{cases} 2\ddot{\theta}_1 + \cos(\theta_1 - \theta_2)\ddot{\theta}_2 + \sin(\theta_1 - \theta_2)\dot{\theta}_2^2 + 2\omega^2\sin(\theta_1) = 0 \\ \ddot{\theta}_2 + \cos(\theta_1 - \theta_2)\ddot{\theta}_1 - \sin(\theta_1 - \theta_2)\dot{\theta}_1^2 + \omega^2\sin(\theta_2) = 0 \end{cases} \quad (20)$$

2.2 数值离散化

考虑数值微分运算

$$\begin{cases} \frac{dy}{dx} \approx \frac{y(x + \Delta x) - y(x - \Delta x)}{2\Delta x} \\ \frac{d^2y}{dx^2} \approx \frac{y(x + \Delta x) + y(x - \Delta x) - 2y(x)}{(\Delta x)^2} \end{cases} \quad (21)$$

令 $\theta_i^{(n)}$ 为第 n 次迭代后 θ_i 的值, 有

$$\begin{cases} \dot{\theta}_1 \approx \frac{\theta_1^{(n+1)} - \theta_1^{(n-1)}}{2\Delta t} \\ \dot{\theta}_2 \approx \frac{\theta_2^{(n+1)} - \theta_2^{(n-1)}}{2\Delta t} \\ \ddot{\theta}_1 \approx \frac{\theta_1^{(n+1)} + \theta_1^{(n-1)} - 2\theta_1^{(n)}}{(\Delta t)^2} \\ \ddot{\theta}_2 \approx \frac{\theta_2^{(n+1)} + \theta_2^{(n-1)} - 2\theta_2^{(n)}}{(\Delta t)^2} \end{cases} \quad (22)$$

令

$$\begin{cases} C_{12} = \cos(\theta_1 - \theta_2) \\ S_{12} = \sin(\theta_1 - \theta_2) \end{cases} \quad (23)$$

回代方程20则有

$$\begin{cases} 2 \frac{\theta_1^{(n+1)} + \theta_1^{(n-1)} - 2\theta_1^{(n)}}{(\Delta t)^2} + C_{12}^{(n)} \frac{\theta_2^{(n+1)} + \theta_2^{(n-1)} - 2\theta_2^{(n)}}{(\Delta t)^2} + S_{12}^{(n)} \left(\frac{\theta_2^{(n+1)} - \theta_2^{(n-1)}}{2\Delta t} \right)^2 + 2\omega^2 \sin(\theta_1^{(n)}) = 0 \\ \frac{\theta_2^{(n+1)} + \theta_2^{(n-1)} - 2\theta_2^{(n)}}{(\Delta t)^2} + C_{12}^{(n)} \frac{\theta_1^{(n+1)} + \theta_1^{(n-1)} - 2\theta_1^{(n)}}{(\Delta t)^2} - S_{12}^{(n)} \left(\frac{\theta_1^{(n+1)} - \theta_1^{(n-1)}}{2\Delta t} \right)^2 + \omega^2 \sin(\theta_2^{(n)}) = 0 \end{cases}$$

即

$$\begin{bmatrix} 0 & S_{12}^{(n)} & 8 & 4C_{12}^{(n)} - 2S_{12}^{(n)}\theta_2^{(n-1)} & c_1 \\ -S_{12}^{(n)} & 0 & 4C_{12}^{(n)} + 2S_{12}^{(n)}\theta_1^{(n-1)} & 4 & c_2 \end{bmatrix} \begin{pmatrix} \left(\theta_1^{(n+1)} \right)^2 \\ \left(\theta_2^{(n+1)} \right)^2 \\ \theta_1^{(n+1)} \\ \theta_2^{(n+1)} \\ 1 \end{pmatrix} = 0 \quad (24)$$

其中

$$\begin{cases} c_1 = 8 \left(\theta_1^{(n-1)} - 2\theta_1^{(n)} \right) + 4C_{12}^{(n)} \left(\theta_2^{(n-1)} - 2\theta_2^{(n)} \right) + S_{12}^{(n)} \left(\theta_2^{(n-1)} \right)^2 + 8(\omega\Delta t)^2 \sin(\theta_1^{(n)}) \\ c_2 = 4 \left(\theta_2^{(n-1)} - 2\theta_2^{(n)} \right) + 4C_{12}^{(n)} \left(\theta_1^{(n-1)} - 2\theta_1^{(n)} \right) - S_{12}^{(n)} \left(\theta_1^{(n-1)} \right)^2 + 4(\omega\Delta t)^2 \sin(\theta_2^{(n)}) \end{cases} \quad (25)$$

该方程需要根据 $S_{12}^{(n)} = 0$ 是否成立分情况讨论。

2.2.1 若考虑二次项系数为零

即 $S_{12}^{(n)} = 0$, 此时有 $\theta_1^{(n)} = \theta_2^{(n)} + k \cdot \pi, k \in \mathbb{Z}$, 此时方程退化为

$$\begin{bmatrix} 8 & 4C_{12}^{(n)} \\ 4C_{12}^{(n)} & 4 \end{bmatrix} \begin{pmatrix} \theta_1^{(n+1)} \\ \theta_2^{(n+1)} \end{pmatrix} = \begin{pmatrix} -c_1 \\ -c_2 \end{pmatrix} \quad (26)$$

其中

$$\begin{cases} C_{12}^{(n)} = \pm 1 \\ (C_{12}^{(n)})^2 = 1 \\ \sin(\theta_2^{(n)}) = C_{12}^{(n)} \sin(\theta_1^{(n)}) \\ \sin(\theta_1^{(n)}) = C_{12}^{(n)} \sin(\theta_2^{(n)}) \\ c_1 = 8(\theta_1^{(n-1)} - 2\theta_1^{(n)}) + 4C_{12}^{(n)}(\theta_2^{(n-1)} - 2\theta_2^{(n)}) + 8(\omega\Delta t)^2 \sin(\theta_1^{(n)}) \\ c_2 = 4(\theta_2^{(n-1)} - 2\theta_2^{(n)}) + 4C_{12}^{(n)}(\theta_1^{(n-1)} - 2\theta_1^{(n)}) + 4(\omega\Delta t)^2 \sin(\theta_2^{(n)}) \end{cases} \quad (27)$$

解得

$$\begin{cases} \theta_1^{(n+1)} &= \frac{1}{4}(C_{12}^{(n)} c_2 - c_1) \\ \theta_2^{(n+1)} &= \frac{1}{4}(C_{12}^{(n)} c_1 - 2c_2) \end{cases} \quad (28)$$

即

$$\begin{cases} \theta_1^{(n+1)} &= \left((2\theta_1^{(n)} - \theta_1^{(n-1)}) + C_{12}^{(n)} (\omega\Delta t)^2 \sin(\theta_2^{(n)}) - 2(\omega\Delta t)^2 \sin(\theta_1^{(n)}) \right) \\ \theta_2^{(n+1)} &= \left((2\theta_2^{(n)} - \theta_2^{(n-1)}) - 2(\omega\Delta t)^2 \sin(\theta_2^{(n)}) + 2C_{12}^{(n)} (\omega\Delta t)^2 \sin(\theta_1^{(n)}) \right) \end{cases} \quad (29)$$

2.2.2 考虑二次项系数不为零

即 $S_{12}^{(n)} \neq 0$, 定义系数

$$\left\{ \begin{array}{l} a = S_{12}^{(n)} \\ b_{11} = \theta_1^{(n-1)} + 2 \cot(\theta_1^{(n)} - \theta_2^{(n)}) \\ b_{12} = \frac{2}{S_{12}^{(n)}} \\ b_{21} = -\frac{4}{S_{12}^{(n)}} \\ b_{22} = \theta_2^{(n-1)} - 2 \cot(\theta_1^{(n)} - \theta_2^{(n)}) \\ c_1 = 8 \left(\theta_1^{(n-1)} - 2\theta_1^{(n)} \right) + 4C_{12}^{(n)} \left(\theta_2^{(n-1)} - 2\theta_2^{(n)} \right) + S_{12}^{(n)} \left(\theta_2^{(n-1)} \right)^2 + 8 (\omega \Delta t)^2 \sin \left(\theta_1^{(n)} \right) \\ c_2 = 4 \left(\theta_2^{(n-1)} - 2\theta_2^{(n)} \right) + 4C_{12}^{(n)} \left(\theta_1^{(n-1)} - 2\theta_1^{(n)} \right) - S_{12}^{(n)} \left(\theta_1^{(n-1)} \right)^2 + 4 (\omega \Delta t)^2 \sin \left(\theta_2^{(n)} \right) \end{array} \right. \quad (30)$$

对于该方程则有

$$\begin{bmatrix} a & 0 & -2ab_{11} & -2ab_{12} & -c_2 \\ 0 & a & -2ab_{21} & -2ab_{22} & c_1 \end{bmatrix} \begin{pmatrix} \left(\theta_1^{(n+1)} \right)^2 \\ \left(\theta_2^{(n+1)} \right)^2 \\ \theta_1^{(n+1)} \\ \theta_2^{(n+1)} \\ 1 \end{pmatrix} = 0 \quad (31)$$

考虑

$$\left\{ \begin{array}{l} \gamma_1 = -\frac{c_2}{a} - b_{11}^2 - 2b_{12}b_{22} \\ \gamma_2 = \frac{c_1}{a} - b_{22}^2 - 2b_{11}b_{21} \end{array} \right. \quad (32)$$

$$\left\{ \begin{array}{l} \theta_{c1}^{(n+1)} = \theta_1^{(n+1)} - b_{11} = \theta_1^{(n+1)} - \left(\theta_1^{(n-1)} + 2 \cot(\theta_1^{(n)} - \theta_2^{(n)}) \right) \\ \theta_{c2}^{(n+1)} = \theta_2^{(n+1)} - b_{22} = \theta_2^{(n+1)} - \left(\theta_2^{(n-1)} - 2 \cot(\theta_1^{(n)} - \theta_2^{(n)}) \right) \end{array} \right. \quad (33)$$

即有

$$\begin{bmatrix} 1 & 0 & 0 & -2b_{12} & \gamma_1 \\ 0 & 1 & -2b_{21} & 0 & \gamma_2 \end{bmatrix} \begin{pmatrix} \left(\theta_{c1}^{(n+1)} \right)^2 \\ \left(\theta_{c2}^{(n+1)} \right)^2 \\ \theta_{c1}^{(n+1)} \\ \theta_{c2}^{(n+1)} \\ 1 \end{pmatrix} = 0 \quad (34)$$

其中 $b_{12} \neq 0$ 与 $b_{21} \neq 0$ 恒成立。通过消元可得

$$\left\{ \begin{array}{l} (\theta_{c1}^{(n+1)})^4 + 2\gamma_1(\theta_{c1}^{(n+1)})^2 - 8b_{12}^2b_{21}\theta_{c1}^{(n+1)} + (4b_{12}^2\gamma_2 + \gamma_1^2) = 0 \\ \theta_{c2}^{(n+1)} = \frac{(\theta_{c1}^{(n+1)})^2 + \gamma_1}{2b_{12}} \end{array} \right. \quad (35)$$

注意到 $b_{21} \equiv -2b_{12}$, 令 $b_{12} = b$, 并换元

$$\begin{cases} p = 2\gamma_1 \\ q = -8b_{12}^2 b_{21} = 16b^3 \\ r = (4b^2\gamma_2 + \gamma_1^2) \end{cases} \quad (36)$$

易得 $\theta_{c1}^{(n+1)}$ 是方程37的其中一根。

$$x^4 + px^2 + qx + r = 0 \quad (37)$$

考虑将该方程转化为的形式

$$(x^2 + \alpha)^2 - (\sqrt{2\alpha - p}x - \frac{q}{2\sqrt{2\alpha - p}})^2 = 0 \quad (38)$$

则系数 α 需要满足

$$\alpha^2 - \frac{q^2}{4(2\alpha - p)} = r \implies 8\alpha^3 - 4p\alpha^2 - 8r\alpha + 4pr - q^2 = 0 \quad (39)$$

而方程的根在复数域则为 ($2\alpha - p < 0$ 同理, 此处不讨论 $\sqrt{-1}$ 与 $\pm i$ 问题)

$$\begin{cases} x_1 = \frac{1}{2}\sqrt{2\alpha - p} + \frac{1}{2}\sqrt{-2\alpha - p - \frac{2q}{\sqrt{2\alpha - p}}} \\ x_2 = \frac{1}{2}\sqrt{2\alpha - p} - \frac{1}{2}\sqrt{-2\alpha - p - \frac{2q}{\sqrt{2\alpha - p}}} \\ x_3 = -\frac{1}{2}\sqrt{2\alpha - p} + \frac{1}{2}\sqrt{-2\alpha - p + \frac{2q}{\sqrt{2\alpha - p}}} \\ x_4 = -\frac{1}{2}\sqrt{2\alpha - p} - \frac{1}{2}\sqrt{-2\alpha - p + \frac{2q}{\sqrt{2\alpha - p}}} \end{cases} \quad (40)$$

考虑到这是该方程的 4 个不重复的根, 考虑到顺序问题可以 α 的实际含义如下

$$\begin{cases} 2\alpha - p &= -(x_1 + x_2)(x_3 + x_4) \\ -\alpha - \frac{1}{2}p + \frac{1}{2}\sqrt{(2\alpha + p)^2 - \frac{4q^2}{2\alpha - p}} &= -(x_1 + x_3)(x_2 + x_4) \\ -\alpha - \frac{1}{2}p - \frac{1}{2}\sqrt{(2\alpha + p)^2 - \frac{4q^2}{2\alpha - p}} &= -(x_1 + x_4)(x_2 + x_3) \end{cases} \quad (41)$$

同一组解可以由 3 个不同的 α 生成, 经验证这些 α 均为方程 39 的解。故仅需得到一个 α 便可求解出所有的根, 故仅考虑方程 39 的实根。

综合方程35、36和方程35、40有

$$\begin{cases} \theta_{c1}^{(n+1)} = \pm_1 \sqrt{\frac{1}{2}(\alpha - \gamma_1)} \pm_2 \sqrt{-\frac{1}{2}(\alpha + \gamma_1) \mp_1 4b^3 \sqrt{\frac{2}{\alpha - \gamma_1}}} \\ \alpha^3 - \gamma_1\alpha^2 - r\alpha + (\gamma_1r - 32b^6) = 0 \\ r = 4b^2\gamma_2 + \gamma_1^2 \end{cases} \quad (42)$$

正负号取决于解的具体形式。

对于 α ，令 $\alpha = \beta + \frac{1}{3}\gamma_1$ 易得

$$\beta^3 - \frac{4}{3}(3b^2\gamma_2 + \gamma_1^2)\beta + \frac{8}{27}(9\gamma_1\gamma_2b^2 + 2\gamma_1^3 - 108b^6) = 0 \quad (43)$$

之前的中间参数 p 、 q 已不再使用，重新定义

$$\begin{cases} p = -\frac{4}{3}(3b^2\gamma_2 + \gamma_1^2) \\ q = \frac{8}{27}(9\gamma_1\gamma_2b^2 + 2\gamma_1^3 - 108b^6) \end{cases} \quad (44)$$

考虑 $\beta = u + v$ 有

$$(u + v)^3 + p(u + v) + q = 0 \implies (u^3 + v^3 + q) + (3uv + p)(u + v) = 0 \quad (45)$$

考虑特解，该解可以构造出一个满足条件 β

$$\begin{cases} u^3 + v^3 = -q \\ uv = -\frac{1}{3}p \end{cases} \quad (46)$$

根据数形结合等方法可以判断 u^3 与 v^3 为方程的两根

$$x^2 + qx - \frac{1}{27}p^3 = 0 \quad (47)$$

即（顺序无关， u 、 v 可以交换）

$$\begin{cases} u = \sqrt[3]{-\frac{1}{2}q + \sqrt{\frac{1}{27}p^3 - \frac{1}{4}q^2}} = \frac{2}{3}\sqrt[3]{\frac{1}{2}\left(-(\dots q) + \sqrt{(\dots q)^2 - 4(\dots p)^3}\right)} \\ v = \sqrt[3]{-\frac{1}{2}q - \sqrt{\frac{1}{27}p^3 - \frac{1}{4}q^2}} = \frac{2}{3}\sqrt[3]{\frac{1}{2}\left(-(\dots q) - \sqrt{(\dots q)^2 - 4(\dots p)^3}\right)} \end{cases} \quad (48)$$

联立方程30、33、35、42、43、48等易得

$$\left\{ \begin{array}{l}
 \theta_1^{(n+1)} = \theta_{c1}^{(n+1)} + \left(\theta_1^{(n-1)} + 2 \cot(\theta_1^{(n)} - \theta_2^{(n)}) \right) \\
 \theta_2^{(n+1)} = \theta_{c2}^{(n+1)} + \left(\theta_2^{(n-1)} - 2 \cot(\theta_1^{(n)} - \theta_2^{(n)}) \right) \\
 \theta_{c2}^{(n+1)} = \frac{\theta_{c1}^2 + \gamma_1}{2b} \\
 \theta_{c1}^{(n+1)} = \pm_1 \sqrt{\frac{1}{2}(\alpha - \gamma_1)} \pm_2 \sqrt{-\frac{1}{2}(\alpha + \gamma_1) \mp_1 4b^3 \sqrt{\frac{2}{\alpha - \gamma_1}}} \\
 \alpha = \frac{1}{3}\gamma_1 + u + v \\
 u = \frac{2}{3} \sqrt[3]{\frac{1}{2}(-\sigma_q + \sqrt{\sigma_q^2 - 4\sigma_p^3})} \\
 v = \frac{2}{3} \sqrt[3]{\frac{1}{2}(-\sigma_q - \sqrt{\sigma_q^2 - 4\sigma_p^3})} \\
 \sigma_p = 3b^2\gamma_2 + \gamma_1^2 \\
 \sigma_q = 9\gamma_1\gamma_2b^2 + 2\gamma_1^3 - 108b^6 \\
 \gamma_1 = -\frac{c_2}{a} - b_{11}^2 - 2bb_{22} \\
 \gamma_2 = \frac{c_1}{a} - b_{22}^2 + 4bb_{11} \\
 a = \sin(\theta_1^{(n)} - \theta_2^{(n)}) \\
 b = 2 \csc(\theta_1^{(n)} - \theta_2^{(n)}) \\
 b_{11} = \theta_1^{(n-1)} + 2 \cot(\theta_1^{(n)} - \theta_2^{(n)}) \\
 b_{22} = \theta_2^{(n-1)} - 2 \cot(\theta_1^{(n)} - \theta_2^{(n)}) \\
 c_1 = 8 \left(\theta_1^{(n-1)} - 2\theta_1^{(n)} \right) + 4C_{12}^{(n)} \left(\theta_2^{(n-1)} - 2\theta_2^{(n)} \right) + S_{12}^{(n)} \left(\theta_2^{(n-1)} \right)^2 + 8(\omega\Delta t)^2 \sin \left(\theta_1^{(n)} \right) \\
 c_2 = 4 \left(\theta_2^{(n-1)} - 2\theta_2^{(n)} \right) + 4C_{12}^{(n)} \left(\theta_1^{(n-1)} - 2\theta_1^{(n)} \right) - S_{12}^{(n)} \left(\theta_1^{(n-1)} \right)^2 + 4(\omega\Delta t)^2 \sin \left(\theta_2^{(n)} \right)
 \end{array} \right. \quad (49)$$

在计算 u 、 v 与 $\theta_{c1}^{(n+1)}$ 时需要考虑虚数与分母为 0 情况

- $\sigma_q^2 - 4\sigma_p^3 < 0$ 的情况

可以证明该情况等价于有 3 个实根的情况

$$\alpha_1, \alpha_2, \alpha_3 = \frac{4}{3} \sqrt{\sigma_p} \cos \left(\frac{1}{3} \arccos \left(-\frac{\sigma_q}{2\sigma_p^{3/2}} \right) + \frac{2\pi k}{3} \right), \quad k = 0, 1, 2 \quad (50)$$

- $\alpha - \gamma_1 = 0$ 的情况：过于复杂难以分析，在模拟过程中通过报错或纯数值解来计算
- $\alpha - \gamma_1 < 0$ 的情况：过于复杂难以分析，在模拟过程中通过报错或纯数值解来计算